



Drugs in Bowel Diseases — MCQ Practice

Pharmacology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A-E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. Glucocorticoids are used in IBD for which purpose?

EASY

- A. Long-term maintenance therapy
- B. Short-term induction of remission in moderate-severe disease
- C. Prevention of relapse only
- D. Treatment of IBS-C specifically
- E. Prophylaxis before colonoscopy

2. Which enzyme should be checked before starting azathioprine or 6-mercaptopurine?

EASY

- A. CYP2D6
- B. TPMT
- C. G6PD
- D. DPD
- E. CYP3A4

3. Which drug class is used for IBS-C (constipation-predominant) but not IBS-D?

EASY

- A. Rifaximin
- B. Linaclotide (guanylate cyclase-C agonist)
- C. Loperamide
- D. Alosetron
- E. Eluxadoline

4. What is the shared mechanism by which sulphasalazine, olsalazine, and balsalazide release their active 5-ASA component?

EASY

- A. Gastric acid hydrolysis
- B. Colonic bacterial cleavage of a prodrug bond
- C. Hepatic CYP450 metabolism
- D. Renal excretion activation
- E. Direct absorption in the stomach

5. A UC patient has been on long-term oral prednisolone for over a year and develops cushingoid features, hyperglycemia, and osteoporosis. What is the underlying prescribing error?

MEDIUM

- A. The steroid dose was too low
- B. Steroids were used for long-term maintenance instead of only short-term induction of remission
- C. Steroids were combined with 5-ASA inappropriately
- D. The steroid was given rectally instead of orally
- E. This is an unavoidable consequence of any steroid use regardless of duration

6. A Crohn's patient with untreated latent TB is started on infliximab without screening. What is the primary risk?

MEDIUM

- A. Pulmonary fibrosis
- B. TB reactivation, since anti-TNF agents carry this recognized risk
- C. Hypertension
- D. Oligospermia
- E. Hemolytic anemia

7. A patient with severe refractory Crohn's disease is started on methotrexate rather than azathioprine. Which statement best reflects the described comparative use of these two immunosuppressants?

MEDIUM

- A. Azathioprine and methotrexate are used interchangeably with no difference in disease preference
- B. Methotrexate is noted as more effective in Crohn's than in ulcerative colitis, which may explain this choice
- C. Methotrexate is only used for IBS, not IBD
- D. Azathioprine has a faster onset than methotrexate, making it always preferred
- E. Methotrexate has no described comparative advantage in Crohn's

8. A patient started on azathioprine two weeks ago is frustrated that their IBD symptoms haven't improved and requests a "faster" alternative. What should they be told about thiopurine onset of action?

MEDIUM

- A. Azathioprine should show effect within days, so an alternative is warranted now
- B. Azathioprine has a delayed onset of action, typically 8-12 weeks, so continued treatment (with monitoring) is expected before assessing efficacy
- C. Azathioprine has no delayed onset and this suggests treatment failure
- D. Azathioprine works immediately in IBD but not in other conditions
- E. The two-week timeframe already exceeds the expected onset window

9. A patient with mild-moderate ulcerative colitis is prescribed mesalamine in a delayed-release formulation rather than plain oral tablets. Why is this formulation choice necessary?

MEDIUM

- A. Mesalamine is destroyed by stomach acid and needs enteric protection only
- B. Mesalamine (pure 5-ASA) is well absorbed in the upper GI tract, so a delayed-release or pH-dependent formulation is needed to ensure it reaches the colon where it acts locally
- C. Mesalamine requires colonic bacteria to be activated, like sulphasalazine
- D. Mesalamine has no absorption anywhere in the GI tract, making formulation irrelevant
- E. Delayed-release formulation is only for taste-masking purposes

10. A patient with IBS-D and coexisting depression is started on a low-dose tricyclic antidepressant (e.g. amitriptyline) rather than an SSRI. What is the rationale for this choice specifically in IBS-D?

MEDIUM

- A. TCAs have no relevant GI effect and the choice is arbitrary
- B. TCAs' constipating side effect works in favor of the diarrhea-predominant subtype, in addition to relieving pain and normalizing motility
- C. SSRIs are contraindicated in all IBS patients
- D. TCAs are chosen only for their antidepressant effect, unrelated to bowel habit
- E. TCAs are specifically indicated only for IBS-C, making this an error

11. A patient is started on vedolizumab for moderate-severe UC after failing anti-TNF therapy. Compared to infliximab, what is the key safety advantage described for vedolizumab's mechanism?

MEDIUM

- A. Vedolizumab has zero infection risk of any kind
- B. Vedolizumab's gut-selective mechanism (blocking lymphocyte trafficking to the gut via $\alpha 4\beta 7$) gives it a safer systemic profile, with rare PML risk lower than natalizumab
- C. Vedolizumab requires no monitoring whatsoever
- D. Vedolizumab is more effective than all anti-TNF agents in every patient
- E. Vedolizumab has no described mechanism difference from anti-TNF drugs

12. A patient with UC is prescribed tofacitinib after failing biologics. A colleague notes this drug would not be appropriate if the patient instead had Crohn's disease alone. Why?

MEDIUM

- A. Tofacitinib is approved for UC only, while upadacitinib is approved for both UC and Crohn's, per the described JAK inhibitor indications
- B. Tofacitinib is contraindicated in all IBD
- C. Tofacitinib only works for IBS, not IBD
- D. Both JAK inhibitors are approved for Crohn's only
- E. This distinction does not exist and the colleague is incorrect

13. A patient taking probiotics for antibiotic-associated diarrhea is admitted to the ICU with a central line and develops fever. What specific caution regarding probiotic use applies here?

MEDIUM

- A. Probiotics have no risk in any hospitalized patient
- B. Probiotics should be used cautiously in immunocompromised, critically ill, or central-line patients due to sepsis risk
- C. Probiotics are only risky in outpatient settings
- D. This fever is unrelated to probiotic use under any circumstance
- E. Probiotics are contraindicated only in pregnancy, not ICU settings

14. A patient with IBS-D fails to respond adequately to loperamide alone. The physician adds rifaximin. What is the rationale for this addition based on rifaximin's described mechanism?

MEDIUM

- A. Rifaximin is a systemic antibiotic with broad absorption
- B. Rifaximin is a gut-selective antibiotic that reduces bacterial overgrowth, and a 2-week course can provide lasting relief in IBS-D
- C. Rifaximin works only for IBS-C
- D. Rifaximin is an opioid agonist like eluxadoline
- E. Rifaximin has no distinct mechanism from loperamide

15. A patient with moderate-severe Crohn's disease is being started on azathioprine. TPMT testing shows intermediate activity (heterozygous). How should this influence dosing strategy, based on the lecture's emphasis on TPMT?

HARD

- A. TPMT results are irrelevant to dosing and can be ignored
- B. Intermediate TPMT activity suggests a need for cautious dose adjustment/closer monitoring, since the lecture emphasizes checking TPMT specifically because low activity causes drug accumulation and myelosuppression — an intermediate result implies intermediate (not absent) risk requiring individualized caution
- C. Intermediate TPMT activity means the standard full dose should be doubled for efficacy
- D. TPMT only matters for methotrexate, not azathioprine
- E. Intermediate activity guarantees no risk of myelosuppression at any dose

16. A patient with UC in remission on mesalamine develops a flare and is escalated to a biologic. The physician chooses ustekinumab over an anti-TNF agent, citing the patient's history of recurrent serious infections. What is the mechanistic basis for expecting a potentially different infection risk profile with ustekinumab compared to anti-TNF therapy?

HARD

- A. Ustekinumab and anti-TNF agents have mechanistically identical infection risk profiles with no meaningful distinction
- B. Ustekinumab blocks the IL-12/23 p40 subunit (inhibiting Th1/Th17 pathways) rather than blocking TNF- α broadly, representing a mechanistically distinct immune target that may carry a different (not necessarily absent) infection risk profile
- C. Ustekinumab has zero infection risk under any circumstance
- D. Ustekinumab works by the same TNF-blocking mechanism as infliximab, just with a different name
- E. This choice has no pharmacological basis and is purely arbitrary

17. A patient is being treated for concurrent IBS-C and mild anxiety. The physician considers an SSRI given its noted utility "when anxiety/depression coexist" with IBS, with the additional note that its motility effect "may suit IBS-C." Contrast this with the TCA rationale for IBS-D described elsewhere in the lecture — what general principle about selecting antidepressants in IBS does this contrast illustrate?

HARD

- A. Antidepressant choice in IBS should be based purely on antidepressant potency, ignoring GI motility effects entirely
- B. The differential motility effects of antidepressant classes (TCAs tending toward constipation, SSRIs tending toward a different motility profile) can be leveraged to match the antidepressant's side-effect profile to the patient's IBS subtype, turning a class side effect into a therapeutic advantage
- C. SSRIs and TCAs have identical motility effects, making this contrast meaningless
- D. Antidepressants should never be used in IBS regardless of psychiatric comorbidity
- E. IBS subtype has no bearing on antidepressant selection

18. A patient with Crohn's disease flare is being managed with the CDED diet approach alongside pharmacotherapy. A dietitian recommends encouraging whole foods, fruits, vegetables, rice, potatoes, lean meats, and eggs, while cutting wheat/gluten grains, dairy, animal fats, and ultra-processed foods. What is the most defensible interpretation of this dietary approach's role relative to pharmacotherapy?

HARD

- A. The CDED diet is presented as a complete replacement for all pharmacologic IBD therapy
- B. The CDED diet is presented as an adjunctive dietary strategy alongside pharmacologic management during a Crohn's flare, not as a replacement for drug therapy such as biologics, immunosuppressants, or steroids
- C. Diet has no described role anywhere in this lecture's IBD management framework
- D. The CDED diet is specifically for IBS-C, not Crohn's disease
- E. The CDED diet replaces the need for TPMT testing before starting azathioprine

19. A patient with acid-peptic disease and coexisting mild-moderate Crohn's disease is on multiple medications. Considering that this lecture groups IBD, IBS, and acid-peptic disorders as three separate, largely non-overlapping drug territories, what does this framework imply about managing a patient with more than one of these conditions simultaneously?

HARD

- A. Because the three drug territories are described as largely non-overlapping, a patient with more than one condition would generally require therapy selected independently from each relevant category (e.g., an IBD-specific agent plus a separate acid-suppressive agent) rather than assuming one drug covers multiple conditions
- B. A single drug class always covers all three conditions simultaneously
- C. IBD and acid-peptic disease are mechanistically identical and require identical drugs
- D. Treating one of the three conditions automatically treats the others
- E. This scenario is impossible since a patient cannot have more than one of these three conditions at once

20. A 45-year-old woman with severe IBS-D has failed loperamide, rifaximin, and eluxadoline, and alosetron is being considered. The lecture notes alosetron is "reserved for severe IBS-D in women who failed other treatments, due to rare serious side effects." What does this restriction pattern suggest about the risk-benefit calculus for alosetron specifically?

HARD

- A. Alosetron has no unique risk profile and the restriction is arbitrary
- B. The restriction to severe, treatment-refractory cases in a specific demographic reflects a risk-benefit calculus where alosetron's efficacy is reserved for patients who have exhausted safer options, given its association with rare but serious adverse effects
- C. Alosetron is actually the safest IBS-D drug and the restriction is a labeling error
- D. This restriction applies equally to loperamide and rifaximin
- E. The restriction is based purely on cost, not safety

✓ Answer key

Q	Answer	Difficulty	Why
1	B	EASY	Glucocorticoids are used short-term to induce remission, never for long-term maintenance.
2	B	EASY	TPMT polymorphisms affect thiopurine toxicity; low activity risks severe myelosuppression.
3	B	EASY	Linaclotide increases fluid secretion and is used for IBS-C, not IBS-D.
4	B	EASY	These three all rely on colonic bacteria to cleave a prodrug bond and release active 5-ASA.
5	B	MEDIUM	Glucocorticoids should only be used short-term to induce remission; the cushingoid picture reflects inappropriate long-term maintenance use.
6	B	MEDIUM	Anti-TNF biologics carry a risk of TB (and hepatitis B) reactivation, requiring pre-treatment screening.
7	B	MEDIUM	Methotrexate is noted as more effective in Crohn's disease than in ulcerative colitis.
8	B	MEDIUM	Thiopurines have a delayed onset of 8-12 weeks; two weeks is far too early to judge efficacy.
9	B	MEDIUM	Unlike the prodrugs that rely on colonic bacterial cleavage, mesalamine (pure 5-ASA) is well absorbed proximally, so it needs delayed-release or pH-dependent delivery to reach the colon intact.
10	B	MEDIUM	Low-dose TCAs relieve pain and normalize motility, and their constipating side effect is specifically useful in IBS-D.
11	B	MEDIUM	

			Vedolizumab's gut-selective $\alpha 4\beta 7$ blockade gives it a safer systemic profile compared to anti-TNF agents, with a rare PML risk lower than natalizumab.
12	A	MEDIUM	Tofacitinib is specifically noted for UC only, while upadacitinib covers both UC and Crohn's.
13	B	MEDIUM	Probiotics require caution in immunocompromised, critically ill, or central-line patients due to sepsis risk.
14	B	MEDIUM	Rifaximin is a gut-selective antibiotic that reduces bacterial overgrowth; a 2-week course can give lasting relief in IBS-D, distinct from loperamide's motility-slowing mechanism.
15	B	HARD	Since TPMT activity determines the rate of thiopurine metabolism and low activity causes toxic accumulation, an intermediate (heterozygous) result implies an intermediate risk profile, warranting cautious dosing and closer monitoring rather than either full-dose treatment or assuming safety.
16	B	HARD	Ustekinumab targets the IL-12/23 p40 subunit (Th1/Th17 pathway inhibition), a mechanistically distinct target from TNF- α blockade; while both carry infection risk, the different immune target provides a physiological rationale for considering it as an alternative pathway in patients with anti-TNF-associated infection concerns.
17	B	HARD	The lecture pairs TCAs' constipating tendency with IBS-D (working in the diarrhea-predominant patient's favor) and notes SSRIs' motility effect may suit IBS-C — illustrating a general principle of matching an antidepressant class's GI side-effect profile to the patient's specific bowel-habit subtype for added benefit.
18	B	HARD	The CDED diet is presented within the IBD section as a dietary approach for Crohn's flares, complementing (not replacing) the described pharmacologic armamentarium (5-ASAs, steroids, immunosuppressants, biologics).
19	A	HARD	Since the lecture explicitly frames IBD, IBS, and acid-peptic disorders as three separate territories that "don't overlap much," a patient with more than one condition would generally need condition-specific therapy selected from each relevant category rather than assuming cross-coverage.
20	B	HARD	The explicit restriction of alosetron to severe, refractory IBS-D in women reflects a deliberate risk-benefit tradeoff — reserving a drug with rare serious side effects for patients who have exhausted safer first-line and second-line options.

Pharmacology GIT Block · MEDucation by Mattin Majid · Source: Dr. Jangi Shawkat Salai lecture slides, SGL 1 (IBD, IBS, acid-peptic).
Practice questions — cross-check every answer against the lecture and summary.